Addis Ababa University

College Of Natural And Computational Science

School Of Information Science

Kidus Shimelis
4th year information system Student

COURSE CURRICULUM

Retrieval-Augmented Generation (RAG) Based AI Assistant

Course Overview

This course provides students with a comprehensive introduction to **Retrieval-Augmented Generation (RAG)**, an advanced approach for developing intelligent AI assistants that generate accurate, reliable, and document-aware responses. The course focuses on integrating **Large Language Models (LLMs)** with **vector database technologies** to enable information retrieval from custom knowledge sources, including PDF files and textual documents.

The curriculum is designed with a strong emphasis on **practical implementation and project-based learning**, ensuring that students acquire hands-on experience in building functional RAG-based systems. The course is structured for **short-term intensive delivery**, balancing theoretical foundations with applied development to meet academic and industry-relevant standards.

Course Objectives

By the end of this course, students will be able to:

- Understand the limitations of standalone LLMs
- Explain the RAG architecture and workflow
- Generate and store embeddings for text data
- Implement vector similarity search
- Build a complete RAG pipeline
- Develop a document-based AI assistant
- Expose the assistant via an API or chat interface
- Demonstrate and explain their system confidently

Course Duration & Structure

- **Total Duration:** 2 Weeks
- **Class Frequency:** 3 Days per Week
- **Total Sessions:** 6
- **Session Length:** 3–4 Hours
- **Delivery Mode:** In-person
- **Teaching Style:** 30% theory, 70% hands-on

Target Audience

- Information System / Computer Science students
- Software engineering trainees
- Backend / full-stack developers
- AI beginners with Python knowledge

Prerequisites

Students should have:

- Basic Python programming knowledge(essential)
- Understanding of APIs (basic)
- Familiarity with Git (recommended)
- **No prior AI or ML experience required**

Technology Stack

- **Programming Language:** Python
- **LLM Framework:** LangChain *or* LlamaIndex
- **Vector Database:** FAISS *or* Chroma
- **Model Provider:** OpenAI or Hugging Face
- **Backend:** FastAPI
- **Frontend:** Simple HTML / JavaScript or React
- **Data Sources:** PDF, TXT documents

DETAILED CURRICULUM BREAKDOWN

WEEK 1: FOUNDATIONS OF RAG

DAY 1: LLM Fundamentals & RAG Introduction

Topics Covered

- Introduction to Artificial Intelligence
- Overview of Large Language Models (LLMs)
- How chatbots work
- Limitations of LLMs (hallucination, outdated knowledge)
- Introduction to Retrieval-Augmented Generation (RAG)
- RAG architecture (Retriever + Generator)
- RAG vs Fine-Tuning

Practical Activities

- Prompt engineering experiments
- Comparing answers with and without retrieval
- Visualizing RAG workflow

Learning Outcomes

Students can:

- Explain what RAG is and why it is needed
- Identify when RAG should be used

DAY 2: Embeddings & Vector Databases

Topics Covered

- What are text embeddings?
- Semantic similarity
- Vector representation of text
- Similarity search concepts
- Introduction to vector databases
- FAISS / Chroma overview

Practical Activities

- Generate embeddings from text
- Store embeddings in vector database
- Perform similarity search using queries
- Tune retrieval using top-k values

Learning Outcomes

Students can:

- Convert text into embeddings
- Retrieve relevant content using vector search

DAY 3: Data Ingestion & First RAG Pipeline

Topics Covered

- Document ingestion (PDF, TXT)
- Text chunking strategies

- Chunk size and overlap
- Metadata usage
- Building the first RAG pipeline
- Context injection into LLM prompts

Practical Activities

- Load PDF documents
- Chunk and embed documents
- Build an end-to-end RAG assistant
- Query documents using natural language

Learning Outcomes

Students can:

- Build a basic working RAG assistant
- Answer questions from uploaded documents

WEEK 2: BUILDING REAL-WORLD ASSISTANTS

DAY 4: Improving RAG Quality & Reliability

Topics Covered

- RAG-specific prompt engineering
- Structuring context effectively
- Reducing hallucinations
- Adding source citations
- Testing answer reliability

Practical Activities

- Improve prompt templates
- Add document references to answers
- Evaluate assistant accuracy

Learning Outcomes

Students can:

- Improve answer quality
- Build more reliable assistants

DAY 5: Backend API & Chat Interface

Topics Covered

- Turning RAG into a backend service
- Introduction to FastAPI
- API request/response flow
- Chat-based user interfaces
- File upload for document ingestion

Practical Activities

- Create a RAG API endpoint
- Build a simple chat interface
- Upload new documents dynamically

Learning Outcomes

Students can:

- Deploy RAG as an API
- Interact with the assistant through a chat UI

DAY 6: Final Project, Demo & Evaluation

Topics Covered

- Final project implementation
- Debugging and optimization
- System explanation and architecture review

Practical Activities

- Independent project development
- Live project demonstrations
- Oral explanation of RAG architecture

Learning Outcomes

Students can:

- Build and present a complete RAG-based assistant
- Explain their system professionally

Final Project

Each student or team must build **one RAG-based assistant**, such as:

- University Notes Assistant
- Policy Document Assistant
- Course FAQ Assistant
- Organization Knowledge Bot

Evaluation Criteria

- Correct RAG Architecture – 40%
- Answer Accuracy & Relevance – 30%
- System Explanation – 20%
- UI & Usability – 10%

Certification Criteria

To receive certification, students must:

- Attend at least 80% of sessions
- Submit a working RAG assistant
- Successfully present their project

Skills Gained

- RAG system design
- Vector database usage
- Prompt engineering for retrieval
- Backend AI integration
- Real-world AI assistant development

Future Extensions (Optional)

- Multi-document RAG
- Memory-enabled assistants
- Hybrid search
- Docker deployment
- Cloud hosting